# Vector Search and Similarity Metrics

Vector similarity search finds embeddings most similar to a query embedding in high-dimensional space. Similarity metrics quantify relatedness between vectors, with different metrics suitable for different scenarios. Cosine similarity measures the angle between vectors, ranging from -1 to 1, with 1 indicating identical direction. Cosine similarity is invariant to vector magnitude, making it ideal for comparing normalized embeddings. Euclidean distance computes straight-line distance between vectors in embedding space, ranging from 0 to infinity. Manhattan distance sums absolute differences along each dimension. Dot product similarity computes the inner product of vectors, equivalent to cosine similarity for normalized vectors. Hamming distance applies to binary vectors, counting differing bits. The choice of similarity metric depends on embedding model training and normalization. Most modern embedding models use normalized embeddings with cosine similarity. Vector databases efficiently compute similarities at scale through approximation algorithms. The search process retrieves k nearest neighbors (k-NN search), returning top-k most similar vectors. Retrieval quality depends on embedding model quality, similarity metric appropriateness, and search parameters. Threshold-based filtering returns only vectors exceeding similarity thresholds. Re-ranking refines results using more expensive similarity metrics or cross-encoders.